Estimated Hours - Project 2 Additions

Scope: Easy, Medium, and Hard AI; Basic Self-Solving mode; and timer/high-score feature.

Calculate Use Case Weight (UUCW)

Use case: "Play Against Easy AI"
Select the mode, reveal a tile, alternate with random AI moves, and view the result.
Complexity: Average | Weight: 10 points

Use case: "Play Against Medium AI"
Select the mode, alternate turns with an AI that uses basic deductions and random fallback, and view the result.
Complexity: Average | Weight: 10 points

Use case: "Play Against Hard AI"
Select the mode, alternate turns with an AI that adds horizontal/vertical 1-2-1 deductions, and view the result.
Complexity: Average | Weight: 10 points

Use case: "Run Basic Self-Solving Mode"
Start the mode, observe automatic random moves, and view the result.
Complexity: Simple | Weight: 5 points

Use case: "Track Game Duration and Best Winning Time"
Display elapsed time, stop at win/loss, save a faster winning time for each mine count, and retain records for later games.
Complexity: Average | Weight: 10 points

Total UUCW = 10 + 10 + 10 + 5 + 10 = 45

Calculate Actor Weight (UAW)

Actor: "Player" (interacts through the graphical interface)
Complexity: Complex | Weight: 3 points
Total UAW = 3

Calculate Unadjusted UCP (UUCP)

UUCP = UUCW + UAW = 45 + 3 = 48

Final Effort

Assume a 15% adjustment (1.15 multiplier) for integrating AI turns, timer lifecycle, and saved high scores with the existing game:

Adjusted UCP = 48 × 1.15 = 55.2

Assume a productivity rate of 3 person-hours per UCP.

Final Estimated Effort = 55.2 × 3 = 165.6 person-hours